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PI Name:	Orasanu, Judith M. Ph.D.		
Project Title:	Enhancing Team Performance for Exploration Missions		
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Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
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Space Biology Special Category:	None		
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Fischer, Ute (Georgia Institute of Technology) Kraft, Norbert (NASA ARC)		
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	Since the beginning of human space flight, incidents relating to the psychological, behavioral, and interpersonal aspects of crew performance have jeopardized crew safety and mission success in both U.S. and Russian programs. Successful long-duration space missions will depend on the ability of crewmembers to collaborate effectively under highly stressful conditions. Interpersonal problems are likely to be exacerbated on long-duration missions, especially as crewmembers become more diverse in terms of culture, gender and professional backgrounds. AIMS: The overall goal of this project is to optimize team psychosocial adaptation and performance in Lunar and Mars missions (Risk #24, NASA Critical Path Roadmap, 2005). Specifically, we seek to enhance the ability of crews to respond adaptively to unanticipated problems, to maintain positive social and task cohesion, and to collaborate effectively under stress. The research will (a) determine the effects of task and interpersonal stressors on team performance and cohesion; (b) identify effective team interaction		

Task Description:	and stress management strategies in a simulated task; (c) determine the effects of individual traits and team composition on team cohesion and performance in a simulation environment; (d) determine individual differences in response to team and task stressors; and (e) develop countermeasures to prevent deterioration of team performance and mitigate the effects of stress. Countermeasures will focus on team and leadership training, coordination between astronauts and mission control, and support for interpersonal and teamwork skills. APPROACH: A teamwork simulation laboratory has been created at NASA Ames Research Center to address the issues above. Teams composed of four or five members engage in an Antarctic or Mars search mission over a period of four days (one day of training, three days involving six simulations). Team and task stressors are manipulated along with team composition (gender and national culture). Time synchronized data (computer-based task performance, physiological measures, audio and video recordings) support analyses of team processes and outcomes, including responses to task and interpersonal stress. Questionnaire data (workload, team dynamics, and individual difference measures, e.g., personality and cognitive processes) provide essential information concerning team composition, leadership, and performance both at the individual and team levels. KEY FINDINGS: During Year One we conducted a study involving all male, all female, or mixed-gender crews (two females and two males). Also, further analyses were performed on data collected in our prior study involving all-male four-person teams. Findings from Year One address (a) factors that affect crew performance and team cohesion and (b) individual differences in stress responses. Simulation mission success was influenced by the extent to which individuals collaborated on major tasks. Collaboration in turn reflected individual personality and team dynamics: Players' levels of neuroticism, agreeableness and conscientiousness influenced individual performance, but only when mediated by team innovation and acceptance and by level of team collaboration. Moreover, personality interacted with task difficulty. Group-level neuroticism was related to performance on difficult scenarios: Higher group neuroticism was associated with poorer performance on difficult search scenarios. Team communication analyses help us to understand how these effects are produced. Successful teams' interactions were more symmetric and inclusive than those of less successful teams. While successful teams gathered more task-critical information than less successful ones, they also expressed more positive affect and less dissent. These communication features may have promoted a positive team climate, which predicted team success. Team cohesion also reflected team task goals (manipulated to affect team stress). Cooperative groups had higher group cohesion and group social efficacy scores than competitive groups. At the individual level, female participants in the competitive condition reported lower group communication efficacy than females in the cooperative condition or males in either condition, indicating that females may be more sensitive to competitive goals. Finally, we examined personality differences in response to task stress from the all-male crews of the baseline study. Analyses showed that personality profiles (as measured by the NEO-FFI) were associated with different physiological stress responses induced by the difficult scenarios. These findings indicate that physiological measures designed to monitor stress during space flight may need to be tailored to individual stress "signatures." IMPLICATIONS: Results from this study will provide (a) technologies for predicting breakdown of team dynamics and performance, (b) guidelines for selecting, training, and assembling teams of astronauts to assure team compatibility and optimize performance, (c) tools for self assessment and team feedback, and (d) strategies for managing stress in multi-cultural and gender-mixed teams. PLANS: Plans for the coming year include two new studies. First, we will complete data collection from multicultural gender-mixed crews in the teamwork simulation. Crews composed of Japanese, Russian and U.S. males and females are being examined to determine how various combinations respond to stressors in terms of teamwork and interpersonal interactions. The second study will examine the impact of a potential countermeasure, interpersonal training protocols, on teamwork and task success. U.S. gender mixed teams will be used in this study. In addition, further analyses will be conducted on data already collected.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Most work in government and industry presently is conducted in teams, at least a significant portion of the time. Better understanding of factors that contribute to effective and supportive team performance for space crews will have numerous benefits, especially in high-risk industries, government, and the military. Those benefits include greater efficiency, fewer errors, greater productivity, and less time lost due to interpersonal frictions or stress in the workplace. Our study will indicate which instruments are valid and useful measures of team performance and group cohesion. Understanding how to measure performance, productivity, and cohesion is the first step to being able to predict those factors. Findings on optimal crew composition will be beneficial in the design of industry work teams or any other teams in which members must work closely in a collaborative environment. In addition, our study will yield information on conflict management strategies and validated team training that could be built into education programs in colleges, professional schools, medical training, and preparation for crisis-management teams (e.g., negotiators, firefighters, police, or disaster relief teams). Countermeasures that result from the present project will have broad application to teams that operate under stressful conditions such as flight crews, air traffic controllers, medical personnel, offshore oil workers, or nuclear power operators.
Task Progress:	Three major activities were accomplished during Year One: data collection, data analysis, and pilot testing of an interpersonal management training protocol. Data Collection Two 4-day long studies were conducted addressing the role of team composition on team dynamics, cohesion, and performance in a simulated Antarctic search mission. Study One involved males and females from the U.S.; Study Two included males and females from Japan and Russia as well as the U.S. The gender-mixed U.S. study involved 23 crews, 48 males and 44 females randomly assigned to crews consisting of 4 males, 4 females, or 2 females and 2 males. Task difficulty was varied within crews (3 difficult, 3 moderate scenarios). Team stress was manipulated between crews via goal instructions: cooperate for team success vs. compete for individual success. To date, 4 crews (out of 16) have been run in the multicultural study, all involving 2 males and 2 females. Data collection is ongoing. Data Analyses Three sets of analyses were conducted with data from the U.S. mixed-gender study. Analysis of variance, path modeling, and multilevel modeling were used. 1. We examined the effects of independent and mediating variables on individual and team performance on the simulated search mission. Major independent variables were team and task stress manipulations, as well as individual and group average traits measured by the Big Five Instrument (BFI). 2. We examined the effects of the same independent variables plus team composition on measures of team cohesion and efficacy. 3. We examined the characteristics of individuals preferred by their teams as leaders, using scores on the BFI, the Conflict MODE scale (a measure of preferred strategies for dealing with conflict), and other individual differences measures. Data from our prior grant (12 U.S. male teams) were further analyzed for (a) individual differences in vulnerability to task stress as reflected in physiological responses and (b) differences in communication patterns (structure, content, and relational aspects) associated with more and less successful team performance. Coding schemes were developed to assess (a) strategy planning and adaptive strategy learning and (b) facial expression of affect and stress. These analyses are continuing. Interpersonal Management Skills

	Training During Year Two we will examine the effects of training programs designed to enhance crews' ability to cope with interpersonal stressors and to work effectively as a team. One training intervention involving 25 hours of interpersonal training exercises, didactic instruction, and feedback has been developed and pilot tested. 12 males and 4 females were trained in groups of 8, after which they were divided into groups of 4 that engaged in the Antarctic search simulation. An alternative team training program and filler tasks for the control condition are being developed.
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